

PRACTICAL GUIDE

External connection

Connecting Unity Catalog to ADLS Gen2

This guide connects Unity Catalog to an Azure Data Lake Storage Gen2 account. The connection is made of three objects: an Access Connector, which gives Databricks a managed identity in Azure; a storage credential, which registers that identity in Unity Catalog; and an external location, which links a storage path to the credential.

Before you start, you need:

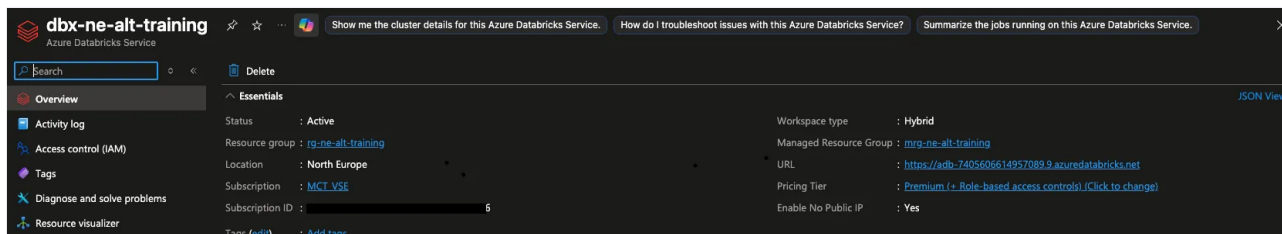
- an Azure Databricks workspace with Unity Catalog enabled
- an Azure Data Lake Storage Gen2 account
- administrator rights in the Azure portal and in the Databricks workspace

Step	Task
1	Copy the resource ID of the Access Connector
2	Grant the Access Connector a role on the storage account
3	Create a storage credential
4	Create an external location
5	Grant access to the external location

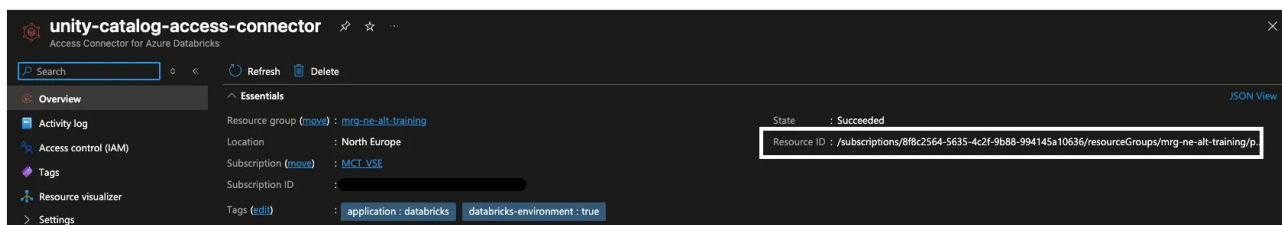
STEP 1

Copy the resource ID of the Access Connector

In the Azure portal, open your Databricks workspace and go to its managed resource group.



The managed resource group contains an **Access Connector for Azure Databricks** resource, named `unity-access-connector` by default. Copy its resource ID; you will need it in step 3.



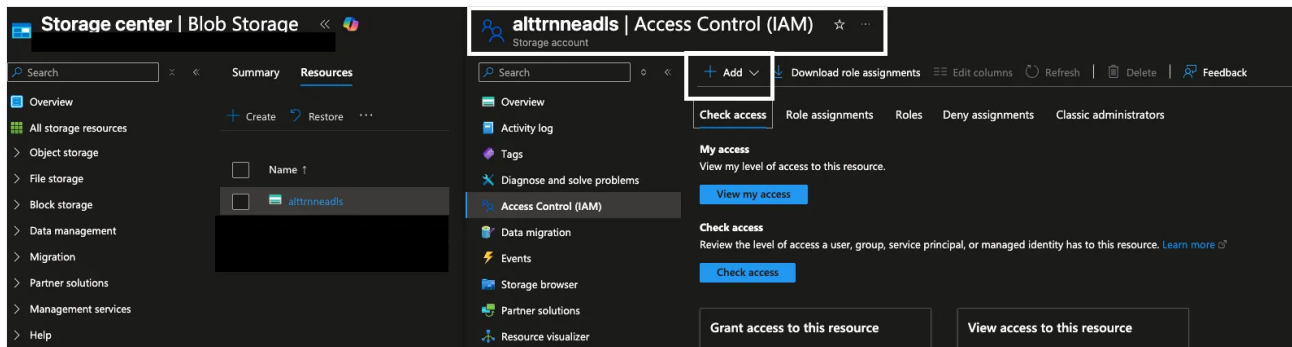
The resource ID has the following form:

```
/subscriptions/<subscription-id>/resourceGroups/<resource-group>/providers/Microsoft.Databricks/
accessConnectors/<name>
```

STEP 2

Grant the Access Connector a role on the storage account

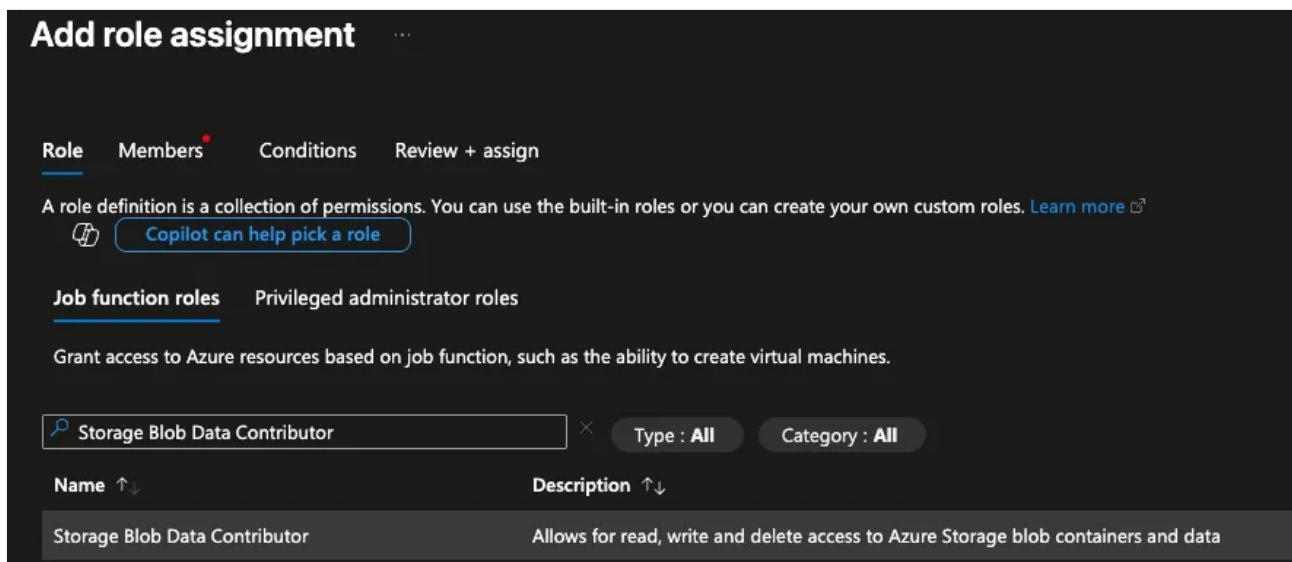
Open the ADLS Gen2 storage account and select **Access control (IAM)**.



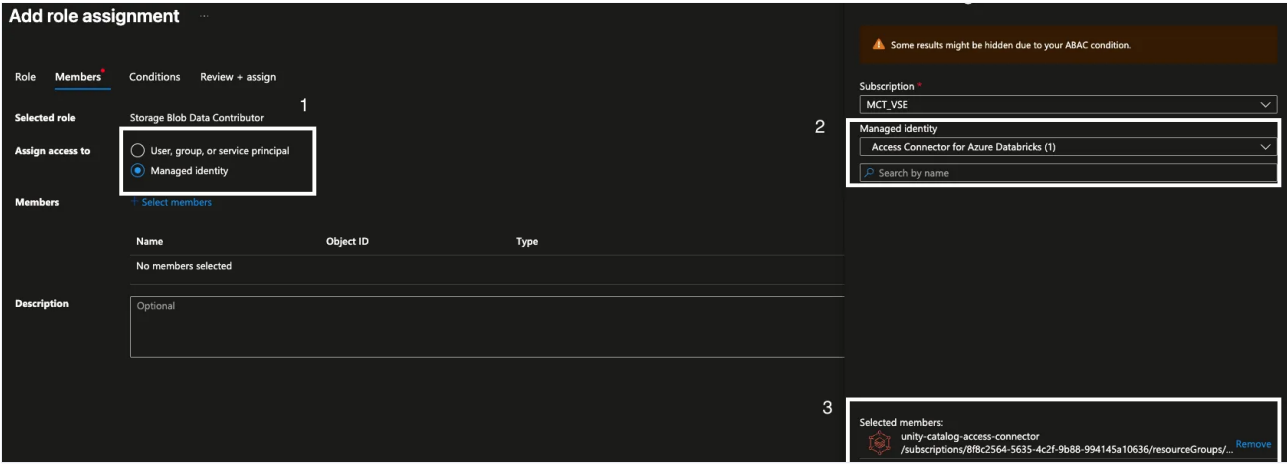
Choose the narrowest role that meets the requirement.

Role	Access	Typical use
Storage Blob Data Reader	Read	Read-only production sources
Storage Blob Data Contributor	Read and write	Training, development and managed storage locations

For this course, assign **Storage Blob Data Contributor**.



On the **Members** tab, choose **Managed identity**, then **Select members**. Filter by **Access Connector for Azure Databricks** and select your connector.

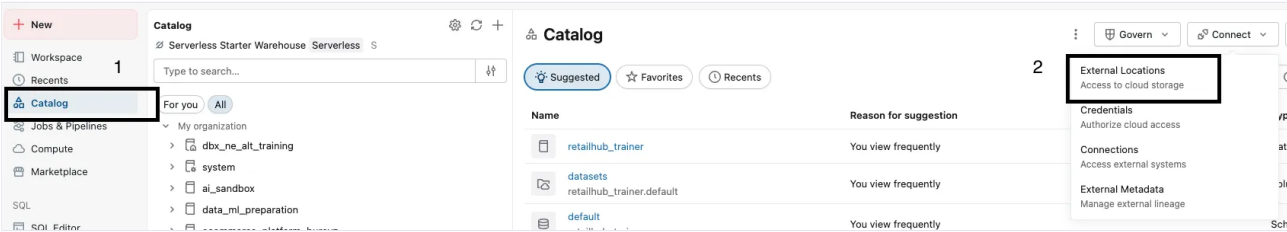


Select **Review + assign**. The Access Connector can now reach the storage account.

STEP 3

Create a storage credential

In the Databricks workspace, open **Catalog**, then **External data** and **Credentials**.



Create a credential, unless a suitable one already exists.

Field	Value
Credential name	A name of your choice, for example <code>adls_training_credential</code>
Access connector ID	The resource ID copied in step 1

One storage credential can be used by several external locations.

STEP 4

Create an external location

Create a dedicated container in the storage account first. Then, on the **External locations** tab, select **Create external location** and complete the form.

Field	Value	Example
External location name	A name of your choice	training_landing_zone
URL	The abfss:// path of the container	abfss://container@account.dfs.core.windows.net/path
Storage credential	The credential created in step 3	adls_training_credential

[External Locations](#) >

Create a new external location

An external location allows you to access your data stored in cloud storage (e.g. Azure Data Lake Storage). You will need the cloud storage path and a paired credential (e.g. managed identity) which gives access to that path [Learn more](#)

External location name*

Storage type*

Azure Data Lake Storage

URL* ⓘ

Enter the bucket path that you want to use as the external location. Note: This must be an ADLS Gen2 storage account with a hierarchical namespace

Copy from DBFS

Storage credential* [Learn more](#)

Provide a storage credential capable of accessing the URL

ext_altrn_storage (Managed Identity)

Connector Id: /subscriptions/8f8c2564-5635-4c2f-9b88-994145a10636/resourceGroups/mrg-ne-alt-training/pro...

User Assigned Managed Identity Id:

Select **Create**. Databricks validates the connection; a result like the one below confirms that the configuration works.

Create a new external location

An external location allows you to access your data stored in cloud storage (e.g. Azure Data Lake Storage). You will need the cloud storage path and a paired credential (e.g. managed identity) which gives access to that path [Learn more](#)

Location Type: Directory

- ✓ Success - Read
- ✓ Success - List
- ✓ Success - Write
- ✓ Success - Delete
- ✓ Success - Path Exists
- ✓ Success - Hierarchical Namespace Enabled

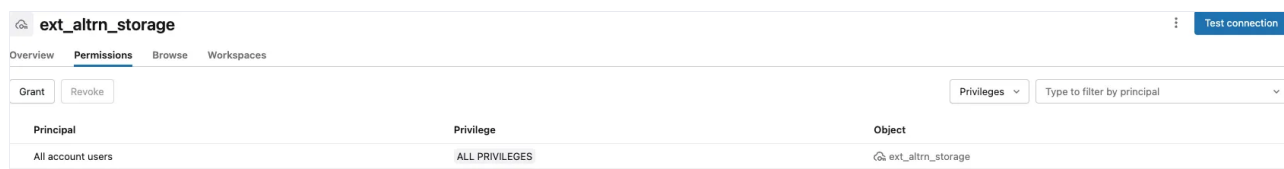
STEP 5

Grant access to the external location

On the **Permissions** tab of the external location, grant privileges to users or groups.

Privilege	Allows
READ FILES	Reading files in the location
WRITE FILES	Writing files to the location
CREATE EXTERNAL TABLE	Creating external tables in the location
ALL PRIVILEGES	All of the above; for development and demonstrations only

The training environment grants **ALL PRIVILEGES** for simplicity. In production, grant each group only the privileges it needs, in line with your organisation's governance policy.



Summary

Where	Step	Result
Azure portal	1. Managed resource group	Resource ID of the Access Connector
Azure portal	2. Storage account, Access control (IAM)	Role assigned to the managed identity
Databricks	3. Catalog, Credentials	Storage credential
Databricks	4. Catalog, External locations	External location with URL and credential
Databricks	5. External location, Permissions	Privileges for users and groups

APPENDIX

Automating the setup with Terraform

The training environment for this course builds the same chain as code, in `infra/terraform/` of the course repository. Terraform creates a dedicated Access Connector instead of reusing `unity-access-connector` from the managed resource group, which keeps the setup reproducible and removable with a single `terraform destroy`.

Step in this guide	Terraform resource	Name in the training environment
1. Access Connector	<code>azurerm_databricks_access_connector</code>	generated from the <code>prefix</code> variable
2. Role on the storage account	<code>azurerm_role_assignment</code>	Storage Blob Data Contributor
3. Storage credential	<code>databricks_storage_credential</code>	<code>sc-<prefix>-training-adls</code>
4. External location	<code>databricks_external_location</code>	<code>el-<prefix>-training-external</code> , container <code>external</code>
5. Grants	<code>databricks_grants</code>	ALL_PRIVILEGES for the training group

Steps 3 to 5 run only when the variable `configure_uc` is `true`, and the identity used for `az login` must be a workspace administrator. Use the manual steps in this guide when you connect a workspace you do not manage with Terraform, and to understand what the code creates.